

L6



CLASS X - SCIENCE

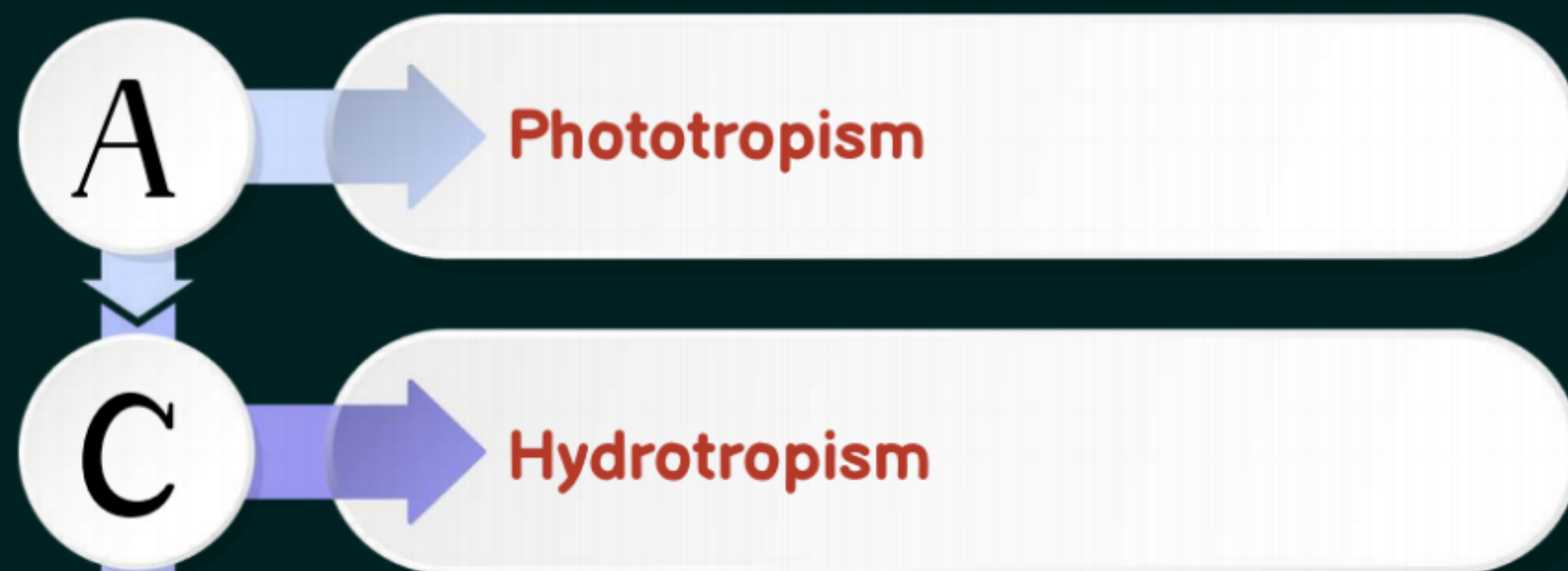
CONTROL AND COORDINATION

| PRASHANT KIRAD |

LET'S TEST YOUR MEMORY



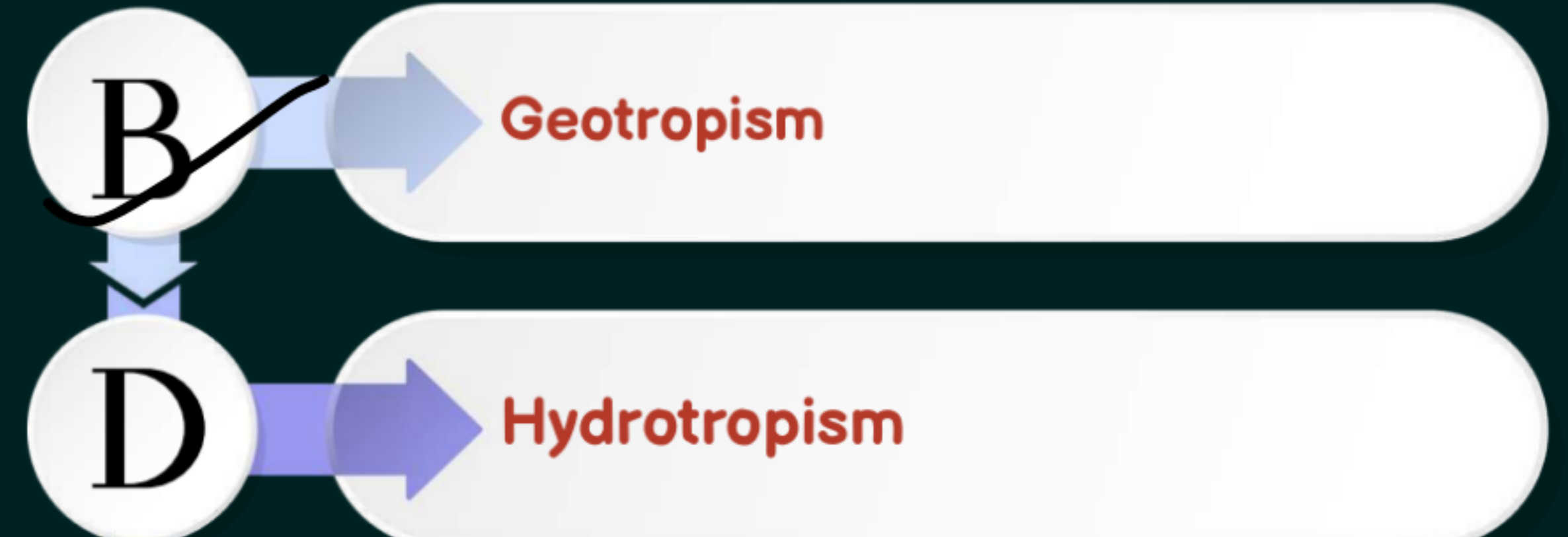
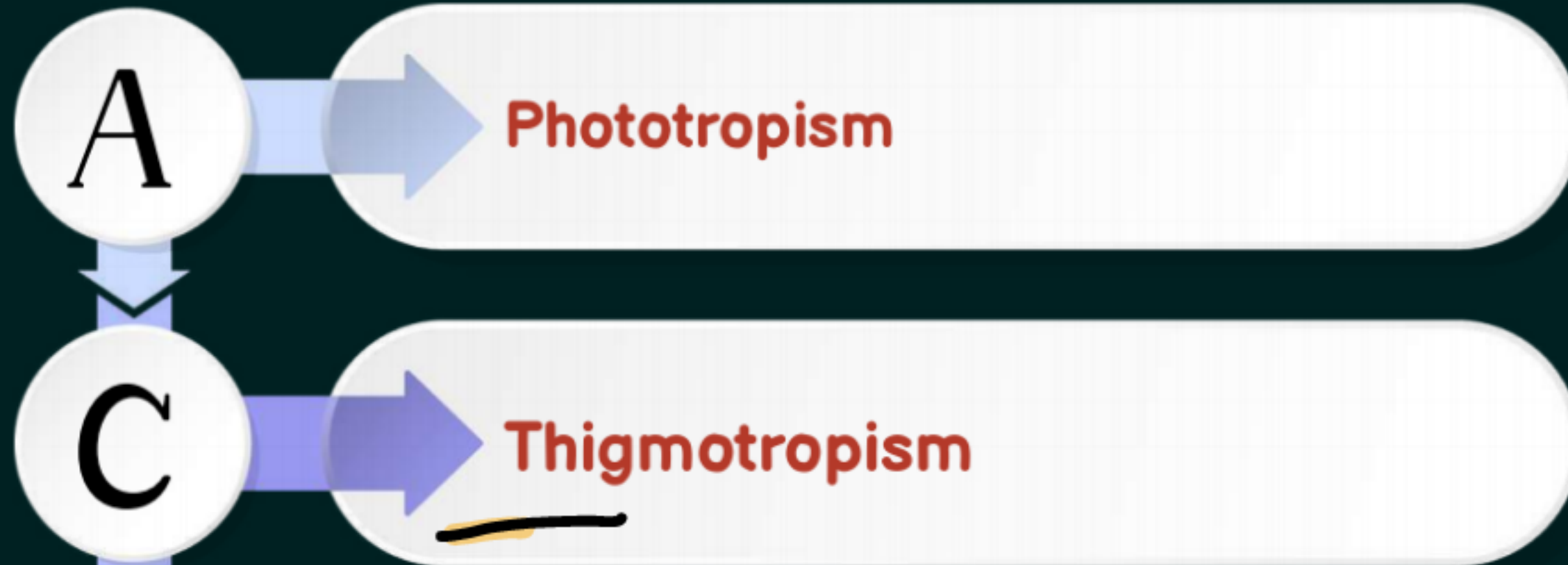
Q. The growth of the pollen tubes towards ovules is an example of:
[CBSE 2025]



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Q. Movement of plant parts in response to gravity is called:[CBSE2021]



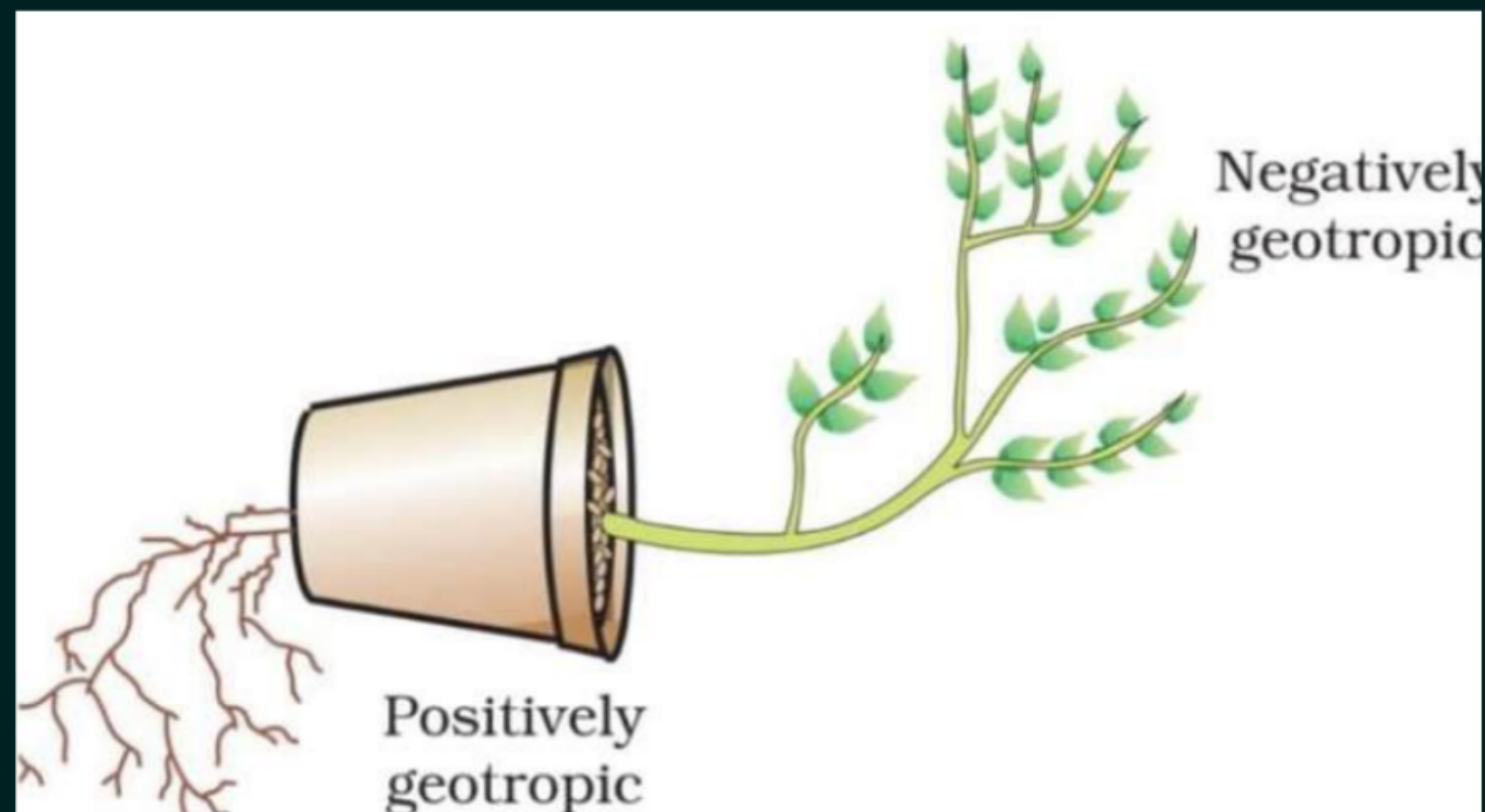
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Q. Study the given diagram and write the type of movement exhibited by: [CBSE 2025]

(a) Root, and

(b) Shoot, mentioning the stimulus in each case.



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Q. Plants have neither a nervous system nor muscles, even then they respond to stimuli. For example, the leaves of chhui-mui (touch-me-not) plant when touched begin to fold up and droop. [CBSE 2025]

- (a) How is the information communicated in "touch-me-not" plants?**
- (b) What enables the plant cells to bring out the observable response?**
- (c) Differentiate the movement mentioned above from the movement of tendrils in a pea plant.**

(a) Information is communicated through electrochemical signals from cell to cell with the help of plant hormones.

(b) Changes in the amount of water in plant cells cause changes in cell shape, leading to folding and drooping of leaves.

(c)

Movement in Touch-me-not Plant	Movement in Tendrils of Pea Plant
It is a nastic movement (thigmonasty).	It is a tropic movement (thigmotropism).
Non-directional response to touch.	Directional growth response to touch.
Not related to growth.	Related to growth.
Movement is fast and temporary.	Movement is slow and permanent.

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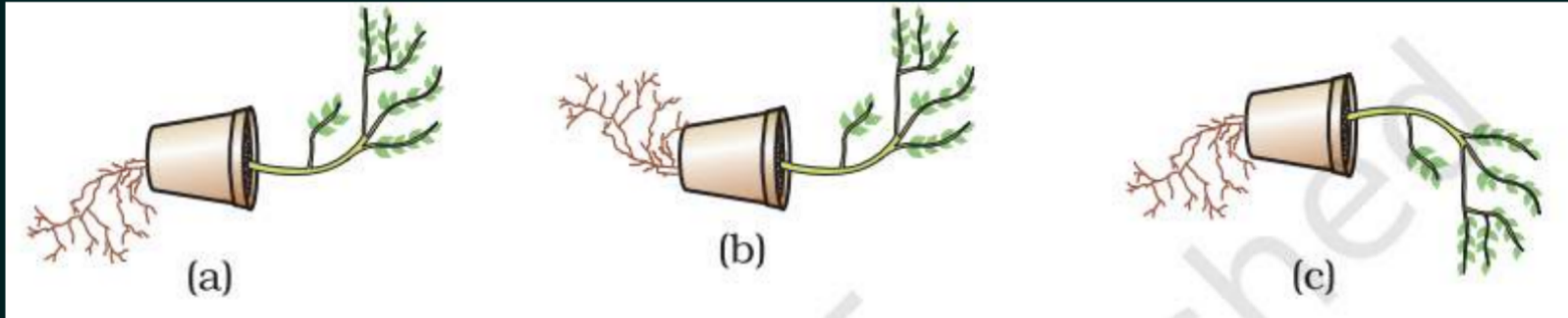
Q. How is the movement of the leaves of a sensitive plant different from the downward movement of the roots? [CBSE 2024].

Movement of Sensitive Plant Leaves	Downward Movement of Roots
It is a nastic movement (<u>thigmonasty</u>).	It is a tropic movement (<u>geotropism</u>).
It is non-directional and does not depend on the direction of stimulus.	It is directional and occurs towards gravity.
It is not related to growth.	It is a growth movement.
It occurs due to changes in water content in cells.	It occurs due to growth of root cells.

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Q. In Figure (a), (b) and (c), which appears more accurate and why?



(A)



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Q. How is the movement of leaves of the sensitive plant different from the movement of a shoot towards light?

Movement of Leaves of Sensitive Plant	Movement of Shoot Towards Light
It is a nastic movement (<u>thigmonasty</u>).	It is a tropic movement (<u>phototropism</u>).
It is non-directional and does not depend on the direction of stimulus.	It is directional and occurs towards light.
It is not related to growth.	It is a growth-related movement.
It occurs due to changes in water content in cells.	It occurs due to unequal growth caused by auxin.
The movement is fast and temporary.	The movement is slow and permanent.

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Q. (i) Leaves of 'chhui-mui' plant begin to fold up and droop in response to a stimulus. Name the stimulus and write the cause for such a rapid movement. Is there any growth involved in the movement?

(ii) Define geotropism in plants. What is meant by positive and negative geotropism? Give one example of each type. [CBSE 2024]

(i) The stimulus is touch.

The rapid movement occurs due to sudden changes in water content (turgor pressure) in the cells of the leaf base.

No, there is no growth involved in this movement. It is a nastic movement.

(ii) Geotropism is the directional growth movement of a plant part in response to gravity.

- Positive geotropism: Growth towards gravity.
 - Example: Roots growing downward.
- Negative geotropism: Growth away from gravity.
 - Example: Shoot/stem growing upward.

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Q. While gardening, Mrs. Mehra noticed that sunflower buds always tilt towards the morning sun but close on cloudy days.

(A) Name the type of plant movement shown in each case.

(B) Explain the role of hormones in the movement towards sunlight.

(C) Why is such a response crucial for the plant's survival?



Answer: (A) ~~Tilting towards the morning sun: Phototropism~~

• Closing on cloudy days: Nastic movement (Photonasty)

(B) The hormone auxin controls phototropism.

Auxin accumulates on the shaded side of the stem, causing those cells to elongate more. This unequal growth bends the plant towards the light.

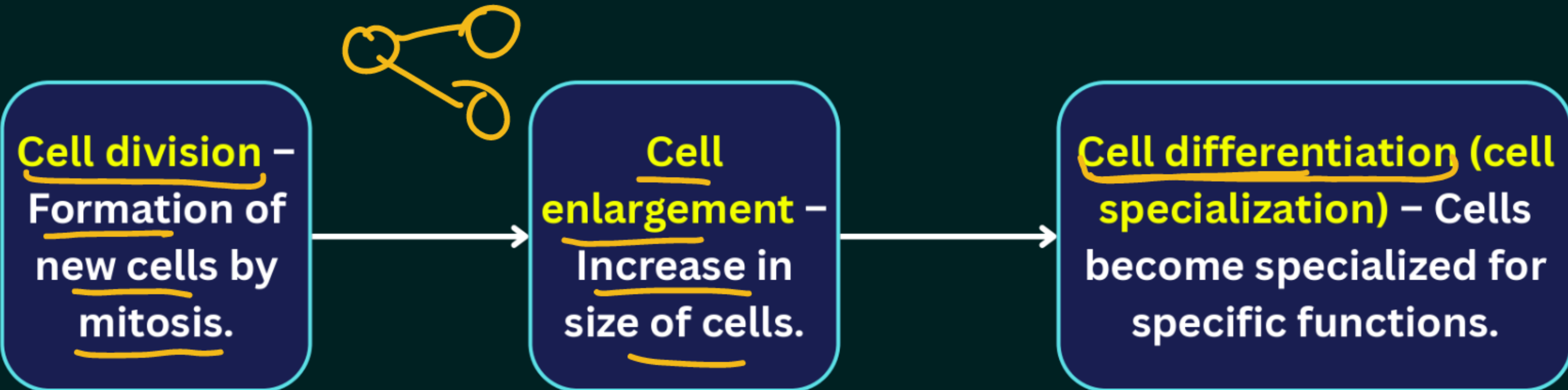
(C) This response helps the plant receive maximum sunlight for photosynthesis, which is necessary for making food and growth.



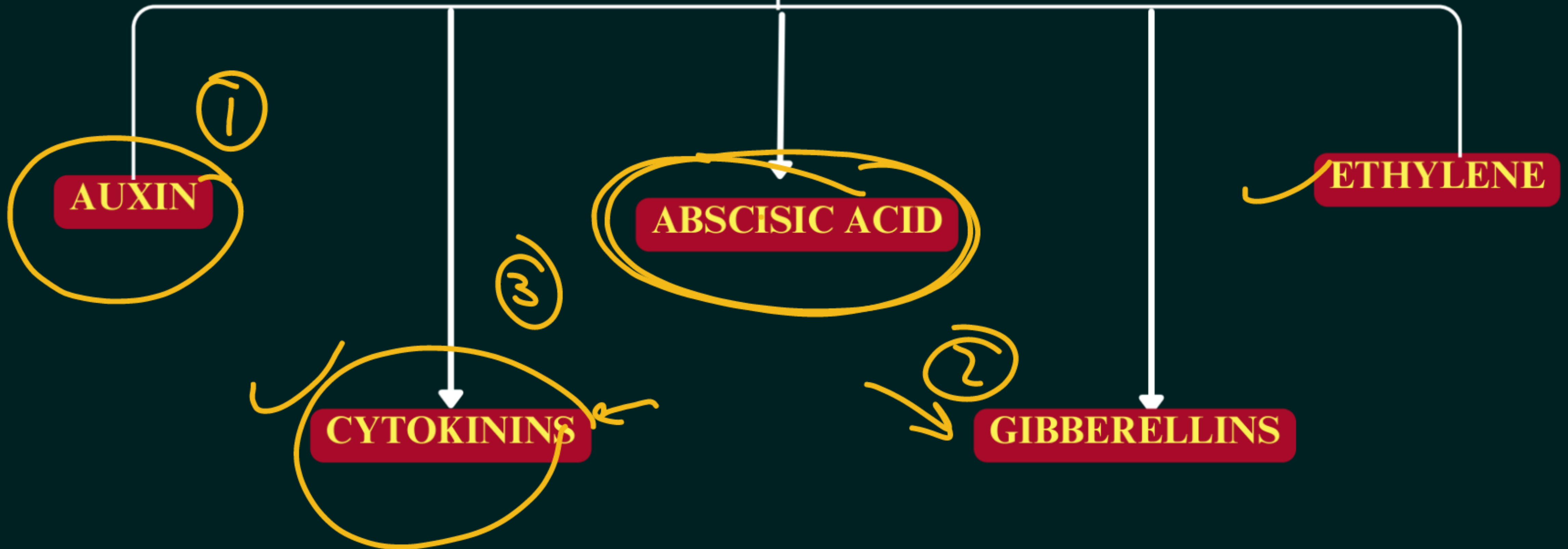
Plant hormones

Plant hormones

- Carried out by plant hormones (phytohormones).
- Growth of a plant occurs in three stages:



Plant hormones

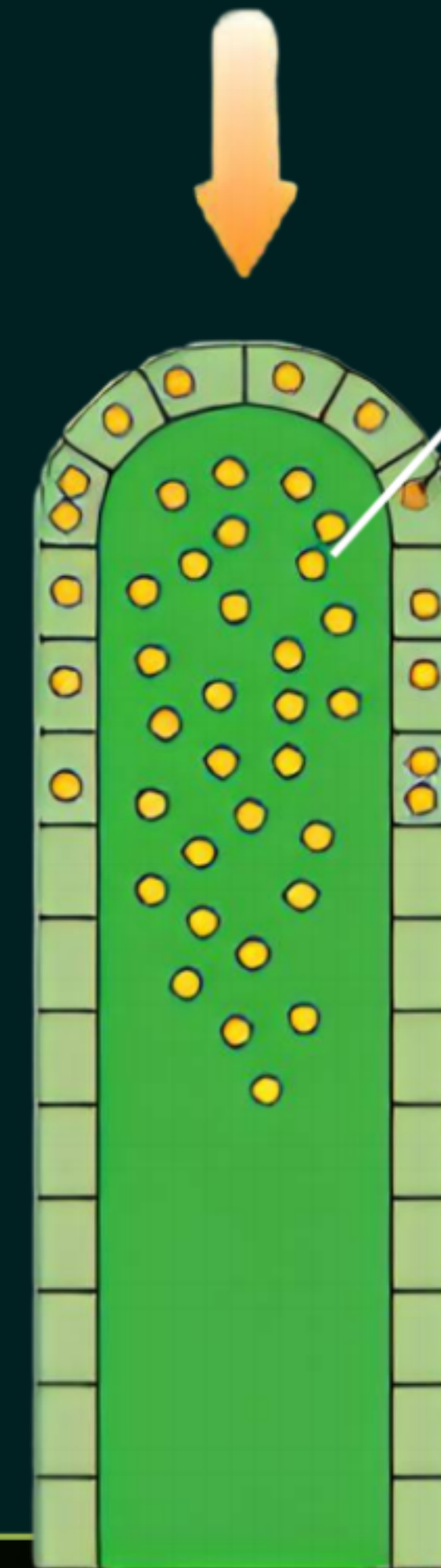


Auxin

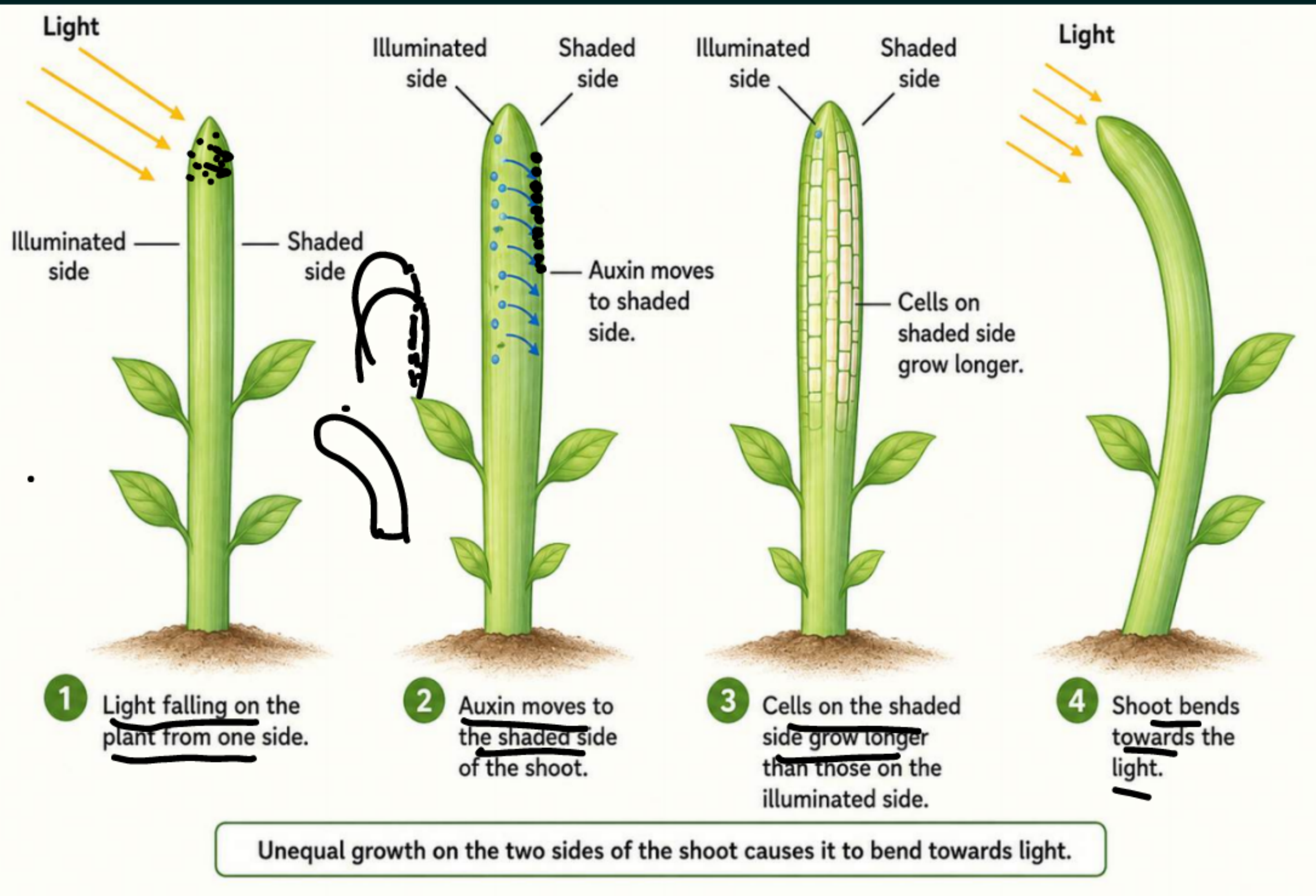
- Auxin is a plant hormone synthesised at the **shoot tip**.
- It helps the cells of the **shoot to grow longer**.
- Auxin plays an important role in the **bending of a shoot towards light**, called **phototropism**.



Auxin Molecules

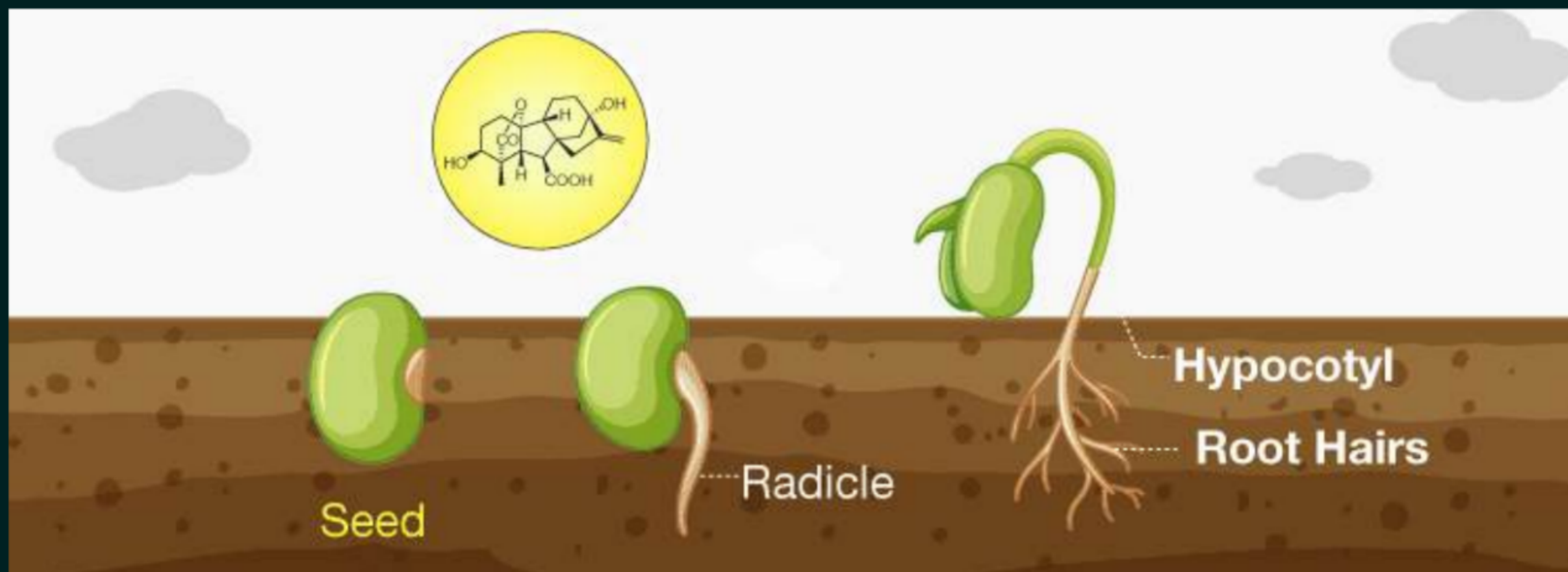


Phototropism in Shoot due to Auxin



Gibberellins

- Gibberellins are **growth-promoting** plant hormones.
- Like auxins, they help in the growth and elongation of the stem.
- Help in seed germination.
- ✓ Promote flowering.
- ✓ Promote fruit development.



Cytokinins

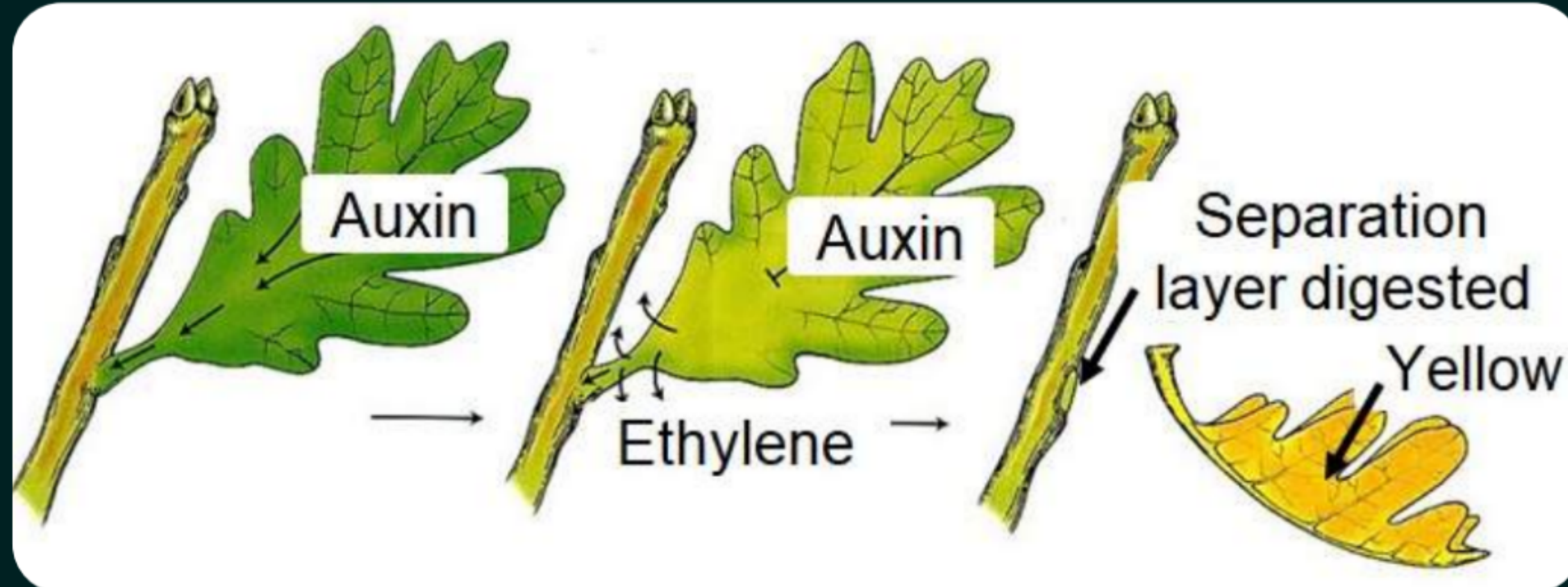
cell division

- Cytokinins **promote cell division.**
- They are present in greater concentration in areas of rapid cell division, such as fruits and seeds.



Abscissic acid (ABA)

- Inhibits growth (opposite of auxin/gibberellins).
- Causes wilting of leaves.
- Induces dormancy in seeds and buds.
- Acts as a stress hormone (helps plant respond to stress - drought).
- Closes stomata during water stress to reduce water loss.



Ethylene → gas

- Stimulates fruit ripening.
- ✓ Promotes leaf fall and aging.

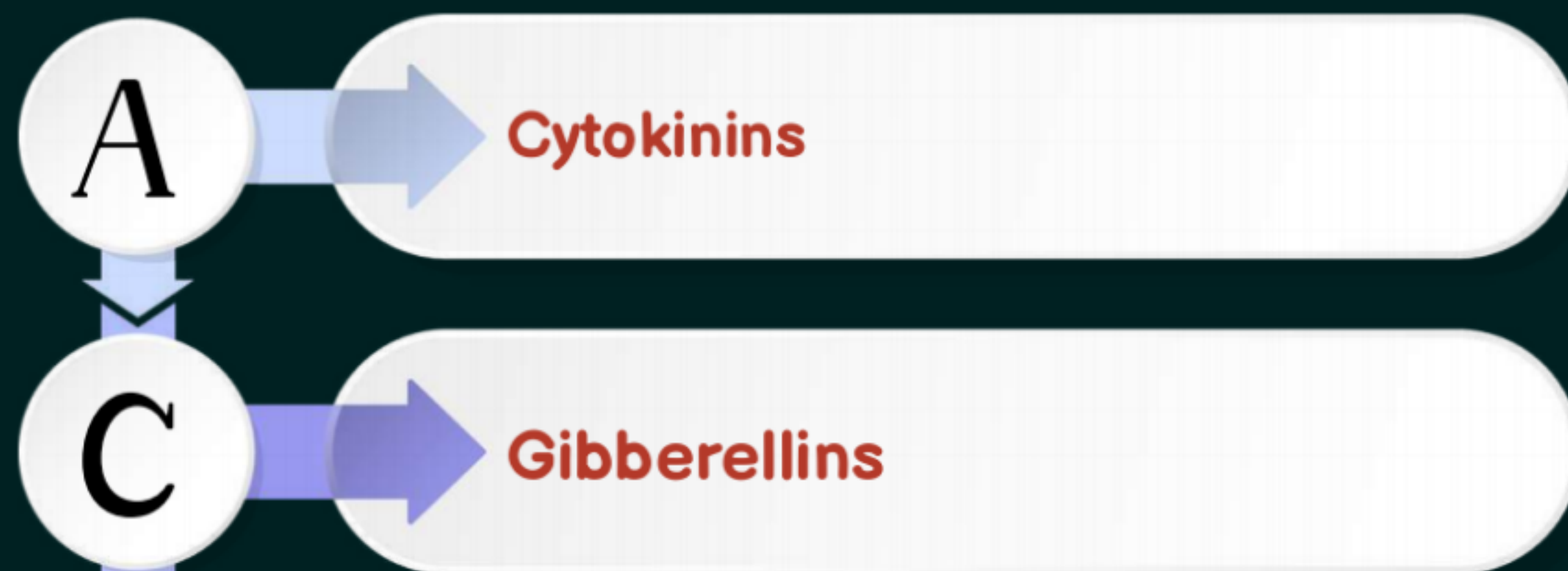


Hormone	Location (Produced at)	Function
<u>Auxins</u>	Shoot tips	Promote cell elongation; control phototropism & geotropism.
<u>Gibberellins</u>	Stem regions	Promote stem elongation; help in seed germination.
<u>Cytokinins</u>	Fruits and seeds (rapid growth areas)	Promote cell division.
<u>Abscisic Acid (ABA)</u>	Various plant parts	Inhibits growth; causes wilting of leaves; acts as a stress hormone.
<u>Ethylene</u>	Fruits (ripening regions)	Stimulates fruit ripening; promotes leaf fall and aging.

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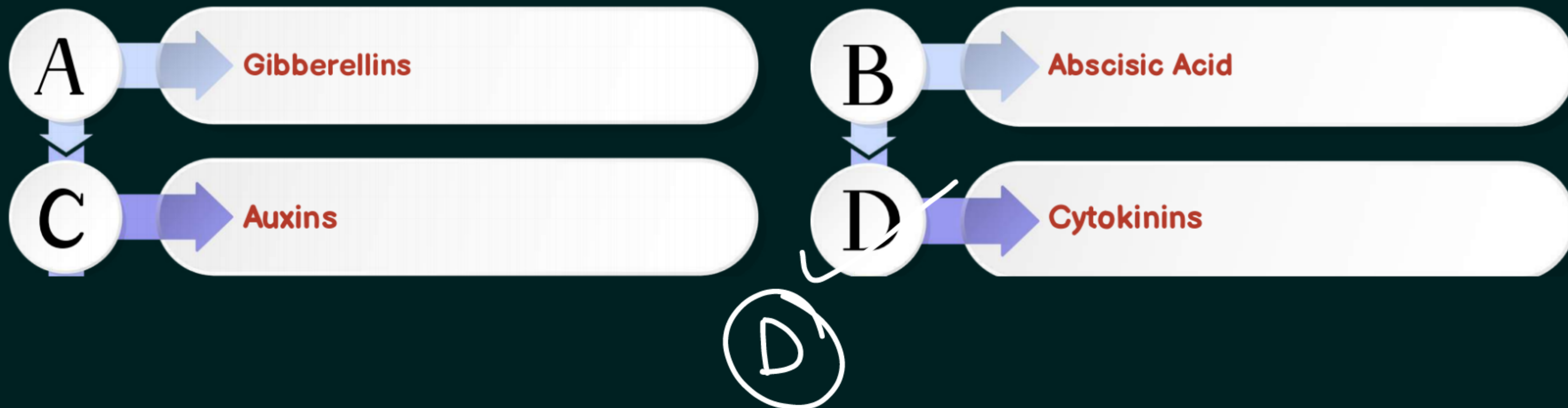
Q. The plant hormone whose concentration stimulates the cells to grow longer on the side of the shoot which is away from light is: [CBSE2025]



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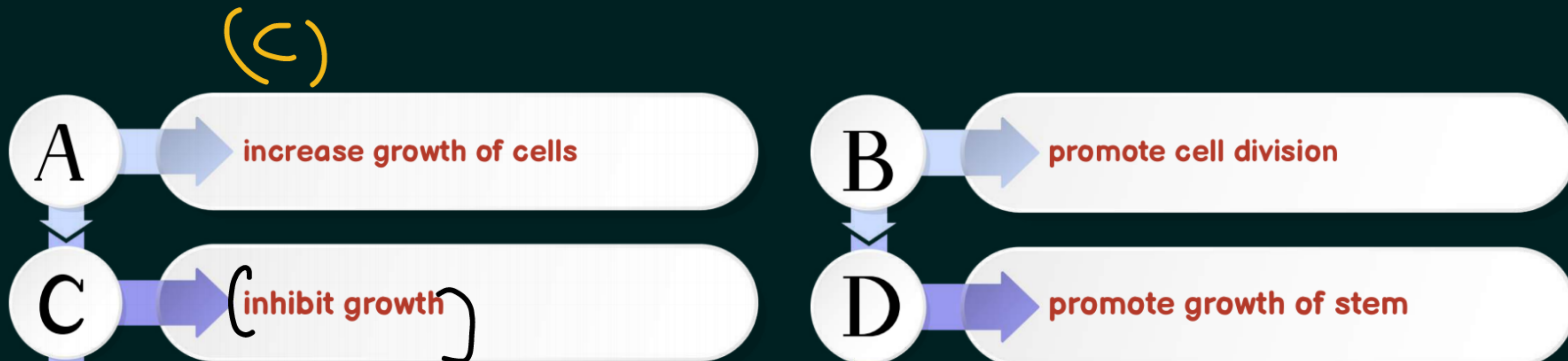
Q. Select from the following a plant hormone which promotes cell division. [CBSE 2024]



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Q. The main function of abscisic acid in plants is to:

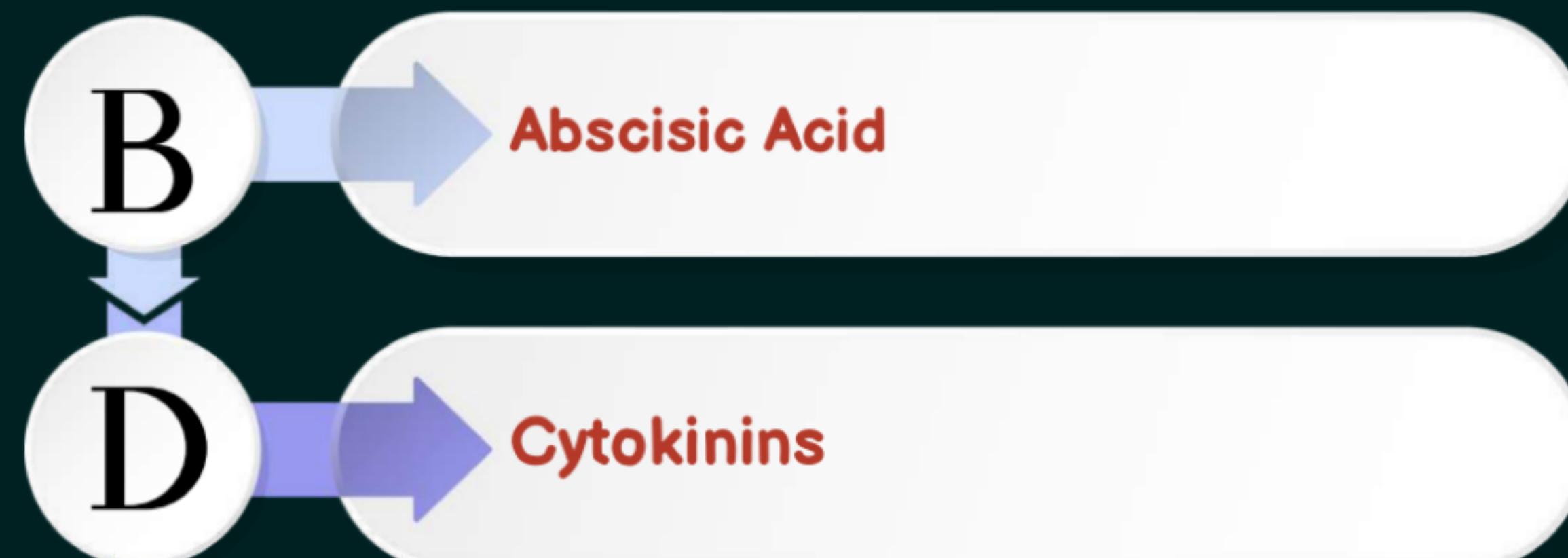
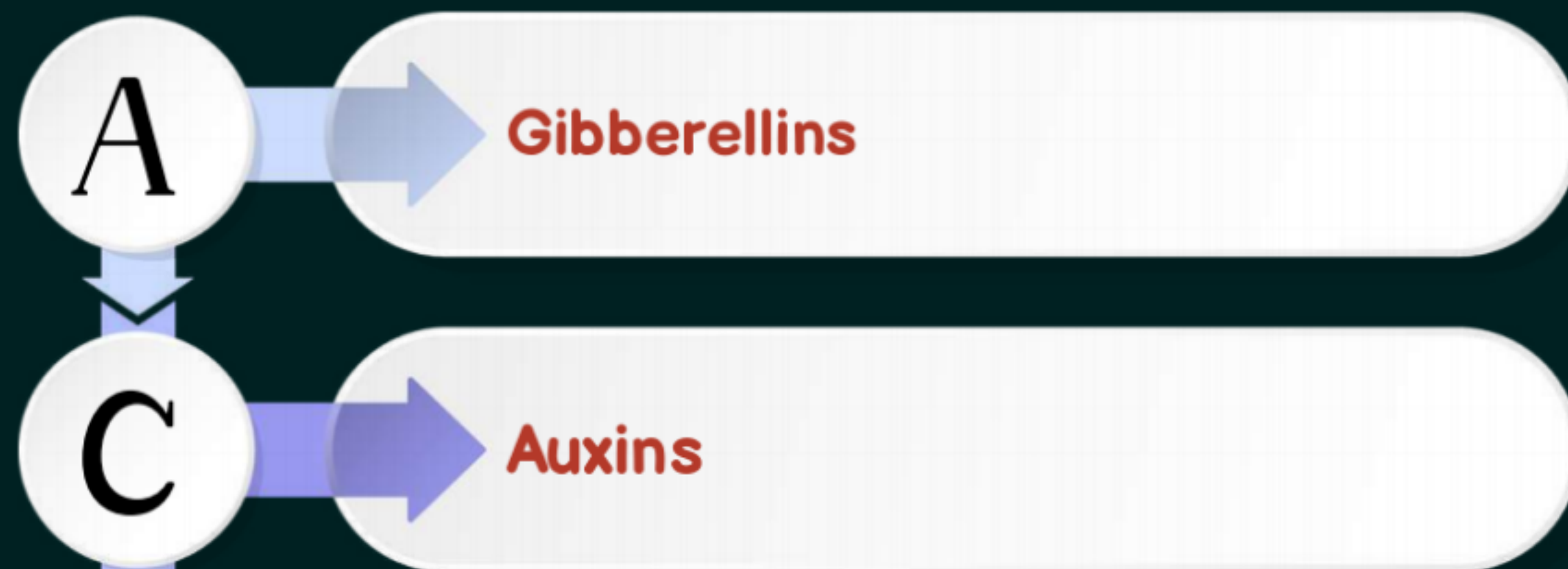


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Q. Which of the following is not associated with growth of plant?

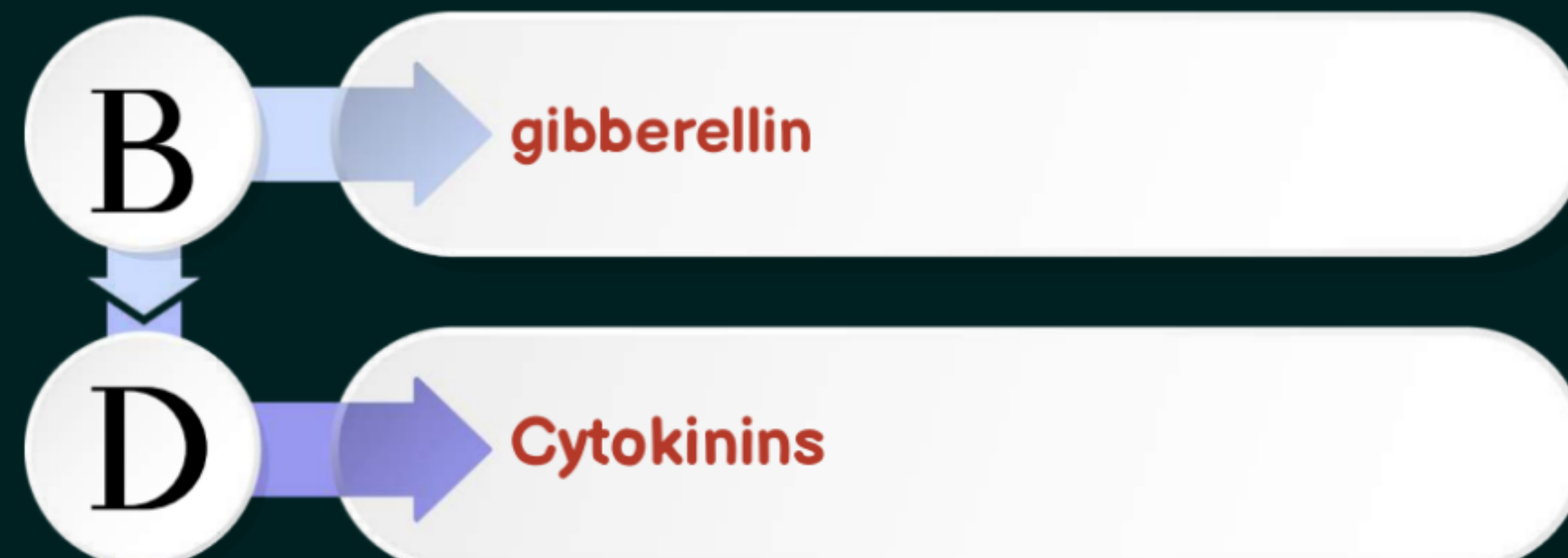
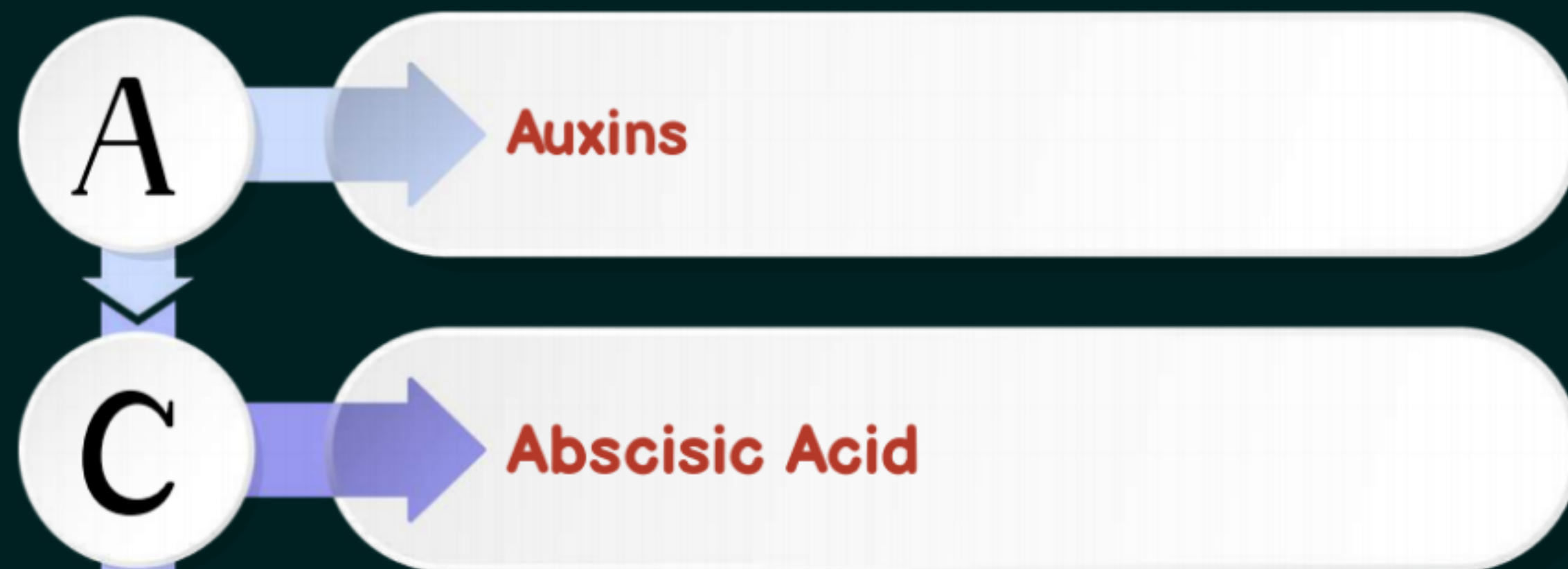
B



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Q. The substance that triggers the fall of mature leaves and fruits from plants is due to?



(c)

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Q. (i) Give an example of a plant hormone that: [CBSE 2024].

(1) inhibits growth and **ABA**

(2) promotes cell division.



(ii) Explain the directional movement of a tendril in a pea plant in response to touch. Name the hormone responsible for this movement.

(i) Tropic movements are directional growth movements of plant parts in response to an external stimulus such as light, gravity, water, or touch.

- Hormone that inhibits growth: Abscissic acid (ABA)
- Hormone that promotes cell division: Cytokinin

(ii) The tendrils of a pea plant show directional growth in response to touch, known as thigmotropism. When a tendril comes in contact with a support, auxin moves to the opposite side of the tendril. Cells on that side grow faster, causing the tendril to bend and coil around the support.

- Hormone responsible: Auxin

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Q. Where are auxins synthesized? How do they promote phototropism?
[CBSE 2023, 2024]

- Auxins are synthesized mainly in the shoot tips and root tips of plants.
- During phototropism, when light falls from one side, auxins move towards the shaded side of the shoot.
- Auxins stimulate greater cell elongation on the shaded side.
- As a result, the shoot bends towards the light, showing positive phototropism.

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Q: Hema bought some unripe tomatoes and left half of them in a brown paper bag and the other half in an open tray. After two days, she noticed that the tomatoes in the paper bag had ripened, but the ones in the open tray had not.

(a) What hormone facilitated the ripening of tomatoes?

(b) Why did the tomatoes in the paper bag ripen faster?

Ans (a) Ethylene

(b) Ethylene gas released by tomatoes accumulated inside the paper bag, increasing its concentration and speeding up ripening.

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Q. Name various plant hormones. Also give their physiological effects on plant growth and development.

Plant Hormone	Physiological Effects on Plant Growth and Development
Auxins ✓.	Promote cell enlargement and cell differentiation; help in phototropism and geotropism; stimulate fruit development.
Gibberellins ✓	Promote stem elongation; break seed and bud dormancy; help in seed germination and fruit growth.
Cytokinins ✓	Promote cell division; delay ageing of leaves; help in seed germination and growth of lateral buds.
Abscissic Acid (ABA) ✓	Acts as a growth inhibitor; causes dormancy in seeds and buds; promotes closing of stomata and falling of leaves.
Ethylene ✓	Promotes fruit ripening; causes wilting and falling of leaves; helps in flowering in some plants.



Bye-bye!

$$\frac{?}{100}$$